

# Accelerating the Deep Learning based Brachial Plexus Nerve Trunks Recognition in Ultrasound Images using Edge Computing

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**Abstract**—The brachial plexus block is an ultrasound image-guided regional anesthesia technique. It involves injecting local anesthetics into or around the brachial plexus to induce numbness in the upper limb, facilitating painless surgeries. Due to the difficulty in achieving clear visualizations of nerves in ultrasound images, this procedure is typically administered by highly trained anesthetists. Therefore, the limited availability of such skilled professionals constrain the widespread performance of these anesthesia procedures. Recent advancements in deep learning-based image segmentation methods, such as U-Net architecture, can potentially solve the segmentation of nerves in ultrasound images. This paper proposes an improved deep learning-based approach to segment brachial plexus nerve trunks in ultrasound images. We improved U-Net architecture by adding a residual block and a naive inception block, achieving a dice coefficient of 0.66. However, the computational complexity of deep learning methods presents a challenge to achieve real-time segmentation using edge computing. As a solution, we deployed the model in Xilinx KV260 multi-processor system on chip (MPSOC) to accelerate the image segmentation.

**Index Terms**—Brachial Plexus, Hardware Acceleration, Deep Learning, Image Segmentation, Medical Image Processing

## I. INTRODUCTION

Brachial Plexus is a nerve network that arises from the spinal cord in the neck and provides motor and sensory innervation to the upper limb of the human body [1]. It is responsible for controlling the muscles and sensations in the shoulder, arm, elbow, wrist, and hand [2]. Highly trained anesthetists perform ultrasound image-guided brachial plexus blocks as a type of regional anesthesia method to numb the upper limb before performing an upper limb surgical procedure. Ultrasound imaging to localization is done in real time, where it often faces several key challenges. One is the in-

adequacy of the anesthetists with extensive experience in handling ultrasound imaging and identifying complex anatomical structures at the same time [3]. Diverse intensity distributions and occlusion caused by factors like acoustic shadows, speckle noise, low contrast, and deformations in anatomical structures make it even more challenging to interpret the ultrasound images [4]. In this procedure, false detection of the nerve trunks may cause nerve injuries. Therefore, this procedure is avoided where general anesthesia can be performed, which is an expensive alternative. Automatic recognition of brachial plexus in ultrasound images can be performed by applying machine learning-based image segmentation methods, and it can assist doctors in performing nerve blocks [5] with higher precision as the accuracy of locating the nerves is directly proportional to the success rate of the nerve block. Yet, due to the complexity of machine learning-based approaches, more computational resources are required to perform segmentation tasks accurately. This work aims to achieve real-time image segmentation of the brachial plexus nerve trunks with a higher accuracy using a hardware accelerator.

Several research studies have focused on addressing the challenging task of identifying the brachial plexus nerve trunks in ultrasound images. These studies have aimed to improve the accuracy of the segmentation task. Recently, deep learning-based image segmentation methods have outperformed traditional algorithms in the domain of image segmentation. U-Net [6] is a widely used deep learning technique in medical image segmentation [7], [8]. U-Net and its variants have performed well for ultrasound brachial plexus nerve segmentation in recent studies. Tian et al. have compared the performance of brachial plexus nerve segmentation with 12 deep learning

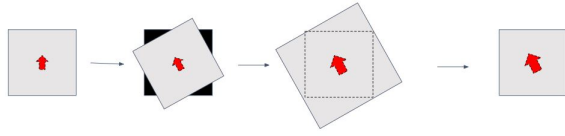


Fig. 1. Shift scale rotate data augmentation method

architectures, and U-Net has delivered the best performance [9]. Wang et al. have shown that adding residual paths to the U-Net model delivers better performance compared to the original U-Net architecture in segmenting brachial plexus region [10].

We present a novel approach to address the challenges linked to the computational resource-intensive nature of deep neural networks, especially in the context of hybrid architectures and expanding datasets with thousands of mathematical operations and large number of weights [11]. This approach utilizes FPGA (Field-Programmable Gate Array) technology, enabling real-time segmentation of the brachial plexus in ultrasound images at the edge. This would be more time efficient and which can protect data privacy and which can even serve the underserved surgical workforce environments.

## II. METHODOLOGY

### A. Data-set and Data pre-processing

The data-set in Kaggle competition under ‘Ultrasound nerve segmentation’ was used to train and test the models [12]. Initially, the data set contains training data with 5635 ultrasound images and their labels. Data-set was split into training and validation data-set in the training phase. Data augmentation techniques were applied to increase the amount of images and the variations in the data-set. The data augmentation technique applied in this scenario is shift scale rotation Fig. 1. The training data-set was increased to 25355 ultrasound images and their labels, and the validation data-set was increased to 2820 ultrasound images and their labels by doing data augmentation.

### B. Model selection and training

Initially, we evaluated two variations of U-Net; ResU-Net [13], and Inception + U-Net [14] models to segment brachial plexus nerve trunk in ultrasound images. Then, we improved the U-Net with residual blocks (ResU-Net) and then crossed with the inception block. The Inception+ResU-Net model in Fig. 2 is capable of recognizing the Brachial Plexus Nerve trunks with the combination of residual block and inception block together with the U-Net model. Residual blocks serve as a skip connection, addressing the problem of vanishing gradients [15] that occurs when gradients become extremely small during backpropagation. The improved model has been enhanced by incorporating inception layers [16] instead of convolution layers in the encoder part of U-Net. The network with inception layers enhances the model’s ability to extract multi-scale features in ultrasound images compared to the

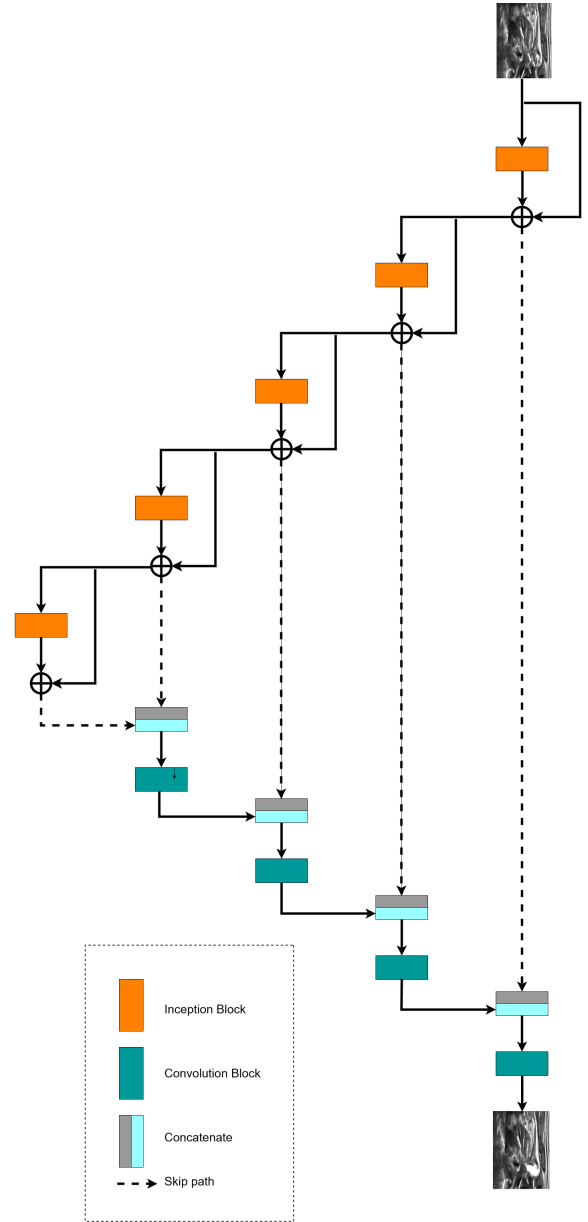


Fig. 2. Inception+ResU-Net architecture

traditional U-Net. The Dice loss function [17] was employed as the loss function during model training.

### C. Hyperparameter optimization

Hyperparameters are tuned in order to achieve better model performance in terms of accuracy, convergence speed, and generalization to unseen data. Learning rate, kernel sizes, and batch size were the optimized hyperparameters. The search space for learning rate was between  $10^{-3}$  and  $10^{-5}$  for kernel sizes of 1,3,5,7,10 and batch sizes of 16 and 32. The optimization was conducted using the random grid search [18], and the combination of hyperparameters that produced

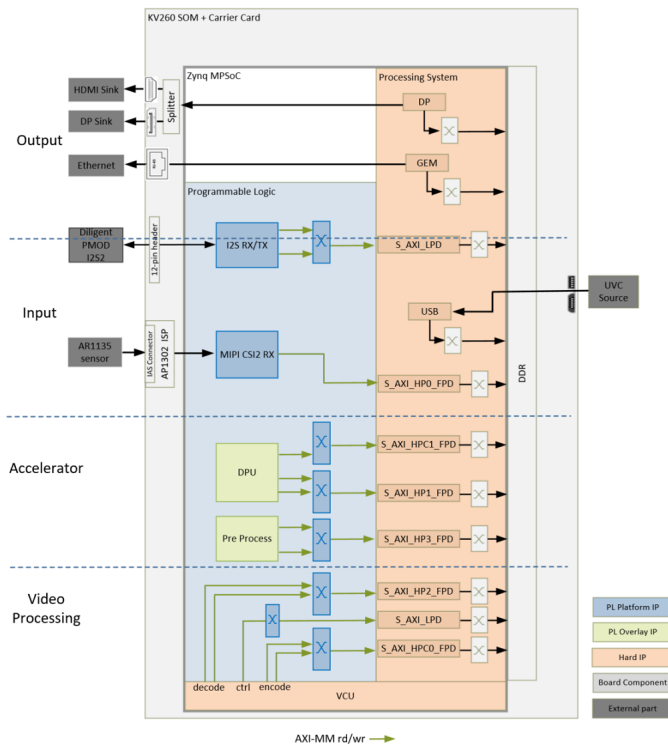


Fig. 3. Hardware architecture of KV260 FPGA [21]

the model with the highest dice coefficient for the validation dataset was identified as optimum hyperparameters.

#### D. Performance evaluation

The Dice coefficient [19], which gives a measure of overlap between the predicted segmentation mask and the ground truth mask was used as a quantitative measure of segmentation accuracy. The evaluation process involved assessing the performance of three distinct models: U-Net, Inception+U-Net, and Inception+ResU-Net, on the validation dataset. The objective was to understand how these models perform when deployed on edge platforms.

### E. Model deployment on the edge

Optimized model in each architecture: U-Net, Inception+U-Net, and Inception+ResU-Net was selected for the inference at the edge. These models were deployed in the selected edge platforms, Intel Core i5-HM370 CPU, NVIDIA Tesla T4 GPU, and Xilinx KV260 MPSOC FPGA. Due to the special architecture of FPGA, several additional steps need to be followed when deploying a neural network. KV260 has a microcode compute engine known as the Deep Processing Unit(DPU) [20], which has an optimized instruction set implemented in the Programmable Logic(PL) of the MPSOC with direct connections to the Processing System(PS) for AI inference. Fig. 3 shows the hardware architecture of KV260 FPGA. Vitis AI is the integrated development environment developed by AMD Xilinx to guide the users in implementing the neural network on DPU of FPGA. The steps include,

- **Inspect the float model:** To understand whether the DPU is compatible with each operation in the neural network. DPU IP used here was DPUCZDX8G IP, which is implemented in the programmable logic (PL) of the selected Zynq® UltraScale+™ MPSoC KV260 board.
- **Quantize the float model:** Vitis AI quantizer [22] reduces the computing complexity while minimizing the loss of prediction accuracy by converting the floating-point weights and activation in the neural networks into fixed-point like INT8. The advantage of this step is providing faster speed and higher power efficiency.
- **Compile the model :** The AMD Vitis AI compiler maps the quantized network model to a highly optimized DPU instruction sequence. Xilinx Intermediate Representation (XIR) is a graph-based intermediate representation of the AI algorithms, which is designed for compilation and efficient deployment of the DPU on the FPGA platform. The compiler is based on XIR, and it takes the quantized tensorflow2 model as the input and transforms it into XIR graph-based format. Then the graph is divided into subgraphs where each operation can be executed on DPU. Then the compiler generates the instruction stream for each subgraph, and then it is serialized into a compiled .xmodel file.
- **Deploy the model :** A system image for KV260 board can be found in the official website [23]. The image is burnt to a SD card and inserted to the board. The subgraphs created through the compiler are available in the model file, and they run in the DPU with python functions calling API for each function.

U-Net, Inception+U-Net and Inception+ResU-Net models are deployed on three edge platforms CPU,GPU and FPGA as mentioned above, and inference results were analyzed to identify the best performing model and the edge platform.

### III. RESULTS AND DISCUSSION

### A. Performance of Machine learning models

All three models are trained and tested with the validation dataset. U-Net achieved a dice coefficient of 0.51, Inception+U-Net 0.55, and Inception+ResU-Net achieved a dice coefficient of 0.60 as shown in Table I. These results were obtained using the models trained in NVIDIA Tesla T4 GPU in the cloud based environment in Google Colab. According to these results, Inception+ResU-Net provides the most accurate predictions for the given dataset. Therefore, Inception+ResU-Net model was selected for further optimization. After the process of hyperparameter optimization, the Inception+ResU-Net model accuracy was incremented from 0.60 to 0.66.

### B. Inference time in different hardware architectures

Optimized Inception+ResU-Net, which showed the highest performance in image segmentation was selected to deploy on edge. To evaluate the inference time, each all the three models U-Net, Inception+U-Net, and optimized Inception+ResU-Net was deployed on different edge devices, Intel Core i5-8300H 4 x 2.3 - 4 GHz Coffee Lake-H Central processing unit(CPU),

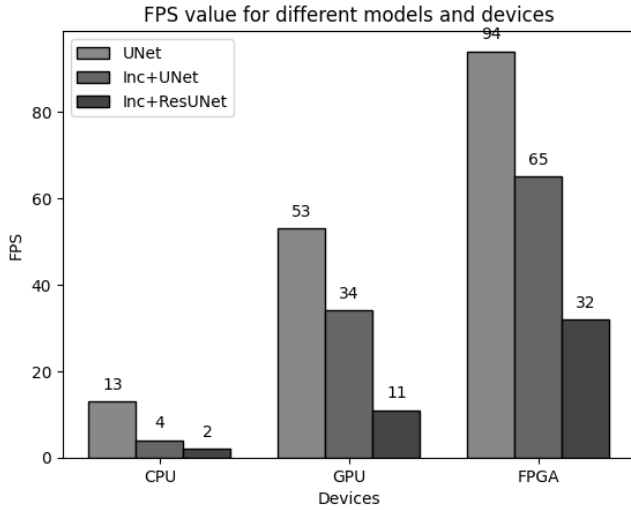


Fig. 4. fps of each model in CPU,GPU and FPGA

NVIDIA Tesla T4 Graphics processing unit(GPU) and Xilinx KV260 FPGA board simultaneously in order to identify which model gives highest accuracy while performing real-time segmentation. The segmentation performance was measured with respect to dice coefficient, and the frames per second(fps) was used as an indicator for the inference time.

According to the results shown in Fig. 4, Inc+ResUNet exhibits a lower fps rate compared to the normal UNet on a specific edge device. This disparity can be attributed to the fact that Inc+ResUNet incorporates additional layers compared to the standard UNet. Consequently, the computational complexity is higher for Inc+ResUNet. Moreover, upon analyzing the findings illustrated in Fig. 4, the fps has increased four times on the GPU compared to the CPU and has increased up to seven times on the FPGA compared to the CPU. When testing the performance with Inception+U-Net model, it is observed that although moving from CPU to GPU and FPGA results in a relatively increased fps, the increment is lower than that observed with the U-Net model. This is attributed to the increased complexity of the Inception+U-Net model compared to the original U-Net model. With increasing complexity, more resources are required, leading to higher inference time. Additionally, the performance of Inception+ResU-Net was evaluated, and it was observed that the fps measured on the FPGA is higher than that on the CPU and GPU. However, it provides a lesser number of frames than both U-Net and Inception+ResU-Net due to its higher complexity. Table I shows the accuracy drop at the quantization process of the FPGA is below 5% for our work which is highlighted in bold.

Most importantly, Fig. 4 emphasizes that, with the introduction of FPGA, the real-time segmentation with 32 fps was achieved for the most accurate model, Inception+ResU-Net. Fig. 5 visualizes the results of the segmentation conducted using the FPGA where the red mask is the prediction and the green mask is the original mask annotated by experienced physicians. It provides clear evidence that our objective to

TABLE I  
DICE COEFFICIENT OF EACH FLOAT MODEL AND QUANTIZED MODEL

| Model                         | Dice Coef.     |           | Reduction of Dice Coef. |
|-------------------------------|----------------|-----------|-------------------------|
|                               | Full precision | Quantized |                         |
| U-Net                         | 0.5107         | 0.4821    | 5.6 %                   |
| Inc+U-Net                     | 0.5543         | 0.5401    | 2.6 %                   |
| <b>Inc+ResU-Net normal</b>    | 0.6029         | 0.5847    | 3 %                     |
| <b>Inc+ResU-Net hyp. tune</b> | 0.6677         | 0.6382    | 4.4 %                   |

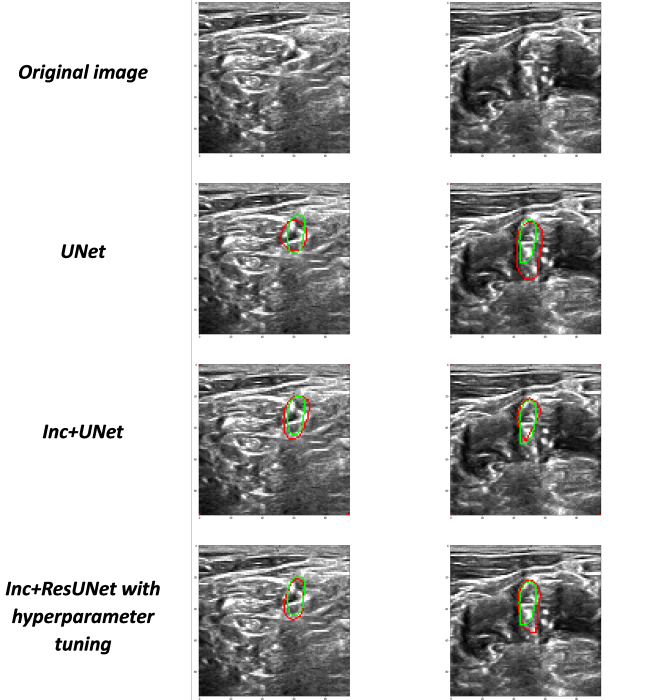


Fig. 5. Segmentation results comparing the original image and ground truth

accelerate the accurate segmentation of brachial plexus nerve trunks in ultrasound images has been achieved.

#### IV. CONCLUSION

We proposed an automated system to segment the nerve trunks in the ultrasound images in real time. The system provides accurate and real-time performance. Processing data using edge computing provides extra security for patients' data. Here, we were able to perform the segmentation task on Xilinx KV260 FPGA successfully. We have achieved a dice coefficient of 66% through a hybrid model architecture of the original U-Net model. We have reached a performance of 32 fps processing ultrasound images, which is a significant improvement compared to previous studies. Due to the low inference time, the proposed system enables real-time ultrasound image-guided brachial plexus nerve recognition to perform nerve blocking.

In this work, key considerations include the selection of an appropriate dataset, the choice of technologies for image segmentation, identifying the most suitable model for segmentation, optimizing the identified model to achieve higher

accuracy, the selection of the optimal edge platform to deploy the model and validation of the system. In conclusion, the Cross U-Net outperforms U-Net and other variants in prediction accuracy. The drawback is the increase in computation complexity and inference time, further exacerbated by network latency. FPGA-accelerated inference is then used to address these limitations in computation complexity and inference time, which are further degraded due to network latency.

The proposed system has many advantages compared to traditional cloud-based systems. It can reduce latency and reliance on the network connection. Furthermore, edge inference helps maintain user privacy since the data never leaves the device. Additionally, with the drastic reduction in cloud resources, edge inference can also reduce ongoing costs. The system's adaptability and generalization to different clinical situations can be achieved by expanding the dataset to include a more diverse range of scenarios and patient profiles. Also, regularly assessing and optimizing the model architecture to accommodate evolving requirements and technological advancements in the field will elevate the system's capabilities.

Future directions beyond this work include extending the application of the developed system to address various medical imaging tasks, exploring its utility in other diagnostic procedures, initiating clinical validation to assess its effectiveness in real-world medical environments, and developing the system as a user-friendly application for seamless integration into medical practices after successful validation.

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